

The 'Harvest Now, Decrypt Later' (HNDL) Threat Model

Day 09 | Category: Foundational PQC Concepts

Date: 2026-07-17 |

DAILY MONITORING

The Urgency of Now: Understanding "Harvest Now, Decrypt Later"

One of the most compelling arguments for the immediate adoption of Post-Quantum Cryptography (PQC) isn't the current existence of a cryptographically relevant quantum computer (CRQC), but a proactive threat model known as **"Harvest Now, Decrypt Later" (HNDL)**. This strategy highlights a significant and immediate risk to long-term data security.

What is HNDL? It is an attack where adversaries intercept and store large volumes of currently encrypted data. They don't have the means to decrypt it today. However, they are banking on the future availability of a powerful quantum computer that will be able to break the underlying classical encryption (like RSA and ECC), thus revealing the stored data years after it was captured.

Why HNDL is a Critical Threat to Internet Security

The HNDL model completely changes the security timeline. It means that any sensitive data transmitted today that needs to remain confidential for more than a few years is already at risk. Consider the following sectors:

- **National Security:** Classified government communications and intelligence data often need to remain secret for decades.
- **Intellectual Property:** Corporate trade secrets, pharmaceutical research, and patented designs have a long shelf life. Their premature exposure could be catastrophic for businesses.
- **Personal Data:** Personally Identifiable Information (PII), health records, and financial data are often stored for long periods due to regulatory requirements.
- **Financial Systems:** Long-term financial contracts and transaction data could be exposed, leading to market manipulation or fraud.

The immediate defense against HNDL in internet protocols like TLS 1.3 is the adoption of **hybrid key exchange** mechanisms. By combining a classical algorithm (like X25519) with a PQC algorithm (like ML-KEM/Kyber), the connection is protected by both. An adversary would need to break *both* the

classical and quantum-resistant algorithms to decrypt the data, making the "decrypt later" part of their plan infeasible.

Security Comparison: Resisting HNDL Attacks

Cryptographic Scheme	Protection Against Classical Attack	Protection Against Quantum Attack	Vulnerability to HNDL
Classical Only (e.g., RSA, ECC)	✔ Strong	✘ Vulnerable	🔴 High: Data harvested today can be decrypted by a future quantum computer.
Hybrid (e.g., X25519 + ML-KEM)	✔ Strong	✔ Strong	🟢 Low: Protected by the PQC layer. Data remains secure even if the classical layer is broken later.
Pure PQC (e.g., ML-KEM only)	✔ Strong	✔ Strong	🟢 Low: Fully resistant to quantum attacks, assuming the algorithm's hardness holds.

Daily Quiz

Test your understanding of the HNDL threat:

1. What is the primary goal of an adversary using the "Harvest Now, Decrypt Later" strategy?
2. Why is hybrid PQC key exchange considered an effective immediate countermeasure to HNDL?
3. Which type of data is most at risk from HNDL attacks?